Data Modelling

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Agenda

- What is Data Modelling
- Terminologies
- Data Modelling Process
- Normalization – 1NF, 2NF, 3NF
- Relationships – 1:m, m:1
- Denormalization
- ER Diagram
- Dimensional Modelling
- Grains, Facts, Dimensions, Snowflake Schema

What is Data Modelling

- Data modelling is the way to define and structure data so that it can cater to the business requirements of the organization.
- Data modelling is the way to tell databases how to store data in tables as per a pre-defined schema.
- It helps identify business entities as one or more tables.
- It defines relationships among tables in databases.

Modelling for Databases

- RDBMS (Relational) & NoSQL (Non-relational)
- Relational Databases
  - OLTP - MySQL, PostgreSQL, Oracle, DB2, SQL Server etc.
  - OLAP (DWH) – Redshift, BigQuery, Teradata, Hive, Synapse etc.
- Non-relational Databases – DynamoDB, MongoDB, Cassandra, Bigtable, Redis, Couchbase, Neptune, Neo4j etc.
- Do all databases need modelling to store data?

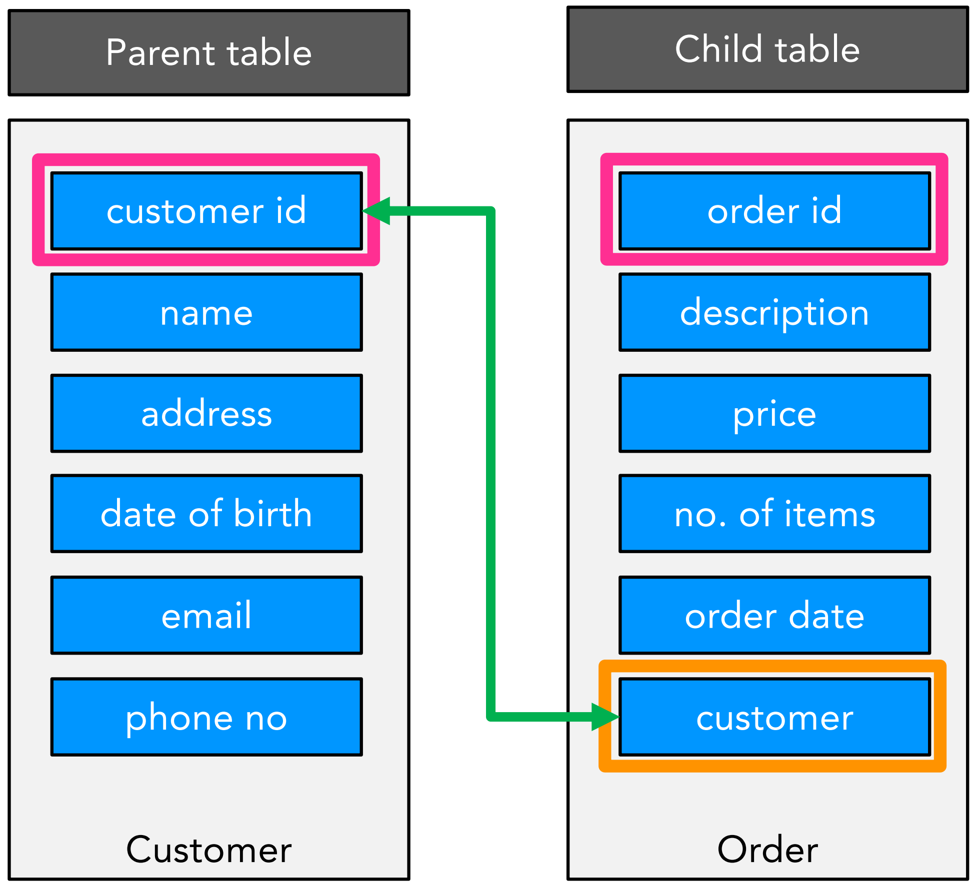
OLTP & OLAP

- OLTP vs OLAP databases.
- OLTP – Modelling technique
  - ER diagram (Entity-Relationship diagram).
  - Normalization – 1NF, 2NF, 3NF.
- OLAP – Modelling technique
  - Dimensional modelling.

OLTP	OLAP
Data is born here.	Data is analyzed here.
Used to run business.	Used to grow business.
Contains operational data.	Contains historical data.
Used for both writes and reads. Some workloads can have higher write percentage.	Used mostly (95%) for reads.
Deals with smaller result sets. 10s to 100s of rows. 1000s of rows seldom.	Deals with very large result sets. 10s of thousands to 100s of thousands of rows.
Handles high concurrency (queries, users, connections).	Very low on concurrency.
Data size varies from 100s of GBs to a few TBs.	Data size can be in 10s of TBs or PBs.
Very fast query processing (milliseconds – seconds).	Query processing time is higher (mins – hours).
Data ingestion is through applications.	Data is ingested from OLTP sources or processing frameworks (Spark, Flink etc.)

Terminologies

- Entities – Business objects represented by tables.
- Attributes – Features that define an entity represented by columns in tables.
- Relationships – Relationship between two entities or tables.
- Primary Key – Used to identify a single row uniquely in one table.
- Foreign Key – Used to define relationship between two or more tables.



Data Modelling Process

- Identify entities.
- Identify relationships between entities.
- Identity the modelling technique.

Modelling Concepts

Normalization

- Data organization technique.
- Used to remove redundancy (data duplication). Why to remove redundancy??
- Used to define data integrity.
- Normal forms:
    - First normal form (1NF).
    - Second normal form (2NF).
    - Third normal form (3NF).
    - Fourth normal form (4NF).
    - Fifth normal form (5NF).
    - Boyce-Codd normal form (BCNF).

1NF

Order id	Order date	Delivery date	Description	Order price	Order items	Customer Id	Customer Name	Customer DOB	Customer email	Address
11024	2024-01-01	2024-01-09	iPhone, Galaxy	999.00, 900.00	2	2201	Vikas	1991-08-01	vks@a.com	Bangalore
11083	2025-02-18	2025-02-11	Lenovo, Boat, IFB	250.00, 83.00, 725.00	2	3812	Jaden	1980-06-05	jdn@b.com	New York
23110	2024-12-23	2024-12-25	Headphone	100.00	1	1209	Amit	1994-10-16	amt@c.com	Chennai

- Define possible Primary Keys (candidate keys).
  - Simple primary key – Contains only one attribute.
  - Composite primary key – Contains one or more attributes.
- A single column cannot have multiple values (repeating group).
- Use multiple rows for those repeating groups. Decomposition.

1NF

Order id	Order date	Delivery date	Description	Order price	Order items	Customer Id	Customer Name	Customer DOB	Customer email	Address
11024	2024-01-01	2024-01-09	iPhone, Galaxy	999.00, 900.00	2	2201	Vikas	1991-08-01	vks@a.com	Bangalore
11083	2025-02-18	2025-02-11	Lenovo, Boat, IFB	250.00, 83.00, 725.00	2	3812	Jaden	1980-06-05	jdn@b.com	New York
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11024	2024-01-01	2024-01-09	Galaxy	900.00	1	2201	Vikas	1991-08-01	vks@a.com	Bangalore
11083	2025-02-18	2025-02-11	Lenovo	250.00	1	3812	Jaden	1980-06-05	jdn@b.com	New York
11083	2025-02-18	2025-02-11	Boat	83.00	1	3812	Jaden	1980-06-05	jdn@b.com	New York
11083	2025-02-18	2025-02-11	IFB	725.00	1	3812	Jaden	1980-06-05	jdn@b.com	New York
23110	2024-12-23	2024-12-25	Headphone	100.00	1	1209	Amit	1994-10-16	amt@c.com	Chennai

1NF

Order id	Order date	Delivery date	Description	Order price	Order items	Customer Id	Customer Name	Customer DOB	Customer email	Address
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11024	2024-01-01	2024-01-09	Galaxy	900.00	1	2201	Vikas	1991-08-01	vks@a.com	Bangalore
11083	2025-02-18	2025-02	[illegible]	[illegible]	[illegible]	[illegible]	Jaden	1980-06-05	jdn@b.com	New York
11083	2025-02-18	2025-02	[illegible]	[illegible]	[illegible]	[illegible]	Jaden	1980-06-05	jdn@b.com	New York
11083	2025-02-18	2025-02-11	IFB	725.00	1	3812	Jaden	1980-06-05	jdn@b.com	New York
23110	2024-12-23	2024-12-25	Headphone	100.00	1	1209	Amit	1994-10-16	amt@c.com	Chennai

Table is in First Normal Form (1NF)
Primary Key – order id + description

2NF

- Table has to be in 1NF.
- The non-key attributes should be fully dependent on primary key candidates (order id + description).
- Non-pk attributes – order date, delivery date, order price, items, customer id, customer name, DOB, email, address

Order id	Order date	Delivery date	Description	Order price	Order items	Customer Id	Customer Name	Customer DOB	Customer email	Address
11024	2024-01-01	2024-01-09	iPhone	999.00	1	2201	Vikas	1991-08-01	vks@a.com	Bangalore
11024	2024-01-01	2024-01-09	Galaxy	900.00	1	2201	Vikas	1991-08-01	vks@a.com	Bangalore
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11024	2024-01-01	2024-01-09	Galaxy	900.00	1	2201	Vikas	1991-08-01	vks@a.com	Bangalore
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11083	2025-02-18	2025-02-11	IFB	725.00	1	3812	Jaden	1980-06-05	jdn@b.com	New York
23110	2024-12-23	2024-12-25	Headphone	100.00	1	1209	Amit	1994-10-16	amt@c.com	Chennai

Order id	Order date	Delivery date	Item Id	Order items	Customer Id
11024	2024-01-01	2024-01-09	100	1	2201
11024	2024-01-01	2024-01-09	200	1	2201
11083	2025-02-18	2025-02-11	300	1	3812
11083	2025-02-18	2025-02-11	400	1	3812
11083	2025-02-18	2025-02-11	500	1	3812
23110	2024-12-23	2024-12-25	600	1	1209

Orders

Child table

Description	Order price	Item Id
iPhone	999.00	100
Galaxy	900.00	200
Lenovo	250.00	300
Boat	83.00	400
IFB	725.00	500
Headphone	100.00	600

Items

Parent table

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Order id	Order date	Delivery date	Description	Order price	Order items	Customer Id	Customer Name	Customer DOB	Customer email	Address
11024	2024-01-01	2024-01-09	iPhone	999.00	1	2201	Vikas	1991-08-01	vks@a.com	Bangalore
11024	2024-01-01	2024-01-09	Galaxy	900.00	1	2201	Vikas	1991-08-01	vks@a.com	Bangalore
11083	2025-02-18	2025-02-11	Lenovo	250.00	1	3812	Jaden	1980-06-05	jdn@b.com	New York
11083	2025-02-18	2025-02-11	Boat	83.00	1	3812	Jaden	1980-06-05	jdn@b.com	New York
11083	2025-02-18	2025-02-11	IFB	725.00	1	3812	Jaden	1980-06-05	jdn@b.com	New York
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Order id	Order date	Delivery date	Description	Order items	Customer Id
11024	2024-01-01	2024-01-09	iPhone	1	2201
11024	2024-01-01	2024-01-09	Galaxy	1	2201
11083	2025-02-18	2025-02-11	Lenovo	1	3812
11083	2025-02-18	2025-02-11	Boat	1	3812
11083	2025-02-18	2025-02-11	IFB	1	3812
23110	2024-12-23	2024-12-25	Headphone	1	1209

Orders

Child table

Customer id	name	DOB	Email	address
2201	Vikas	1991-08-01	vks@a.com	Bangalore
3812	Jaden	1980-06-05	jdn@b.com	New York
1209	Amit	1994-10-16	amt@c.com	Chennai

Customers

Parent table

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Order id	Order date	Delivery date	Description	Order price	Order items	Customer Id	Customer Name	Customer DOB	Customer email	Address
11024	2024-01-01	2024-01-09	iPhone	999.00	1	2201	Vikas	1991-08-01	vks@a.com	Bangalore
11024	2024-01-01	2024-01-09	Galaxy	900.00	1	2201	Vikas	1991-08-01	vks@a.com	Bangalore
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11083	2025-02-18	2025-02-11	Boat	83.00	1	3812	Jaden	1980-06-05	jdn@b.com	New York
11083	2025-02-18	2025						1980-06-05	jdn@b.com	New York
23110	2024-12-23	2024						1994-10-16	amt@c.com	Chennai

Tables are in Second Normal Form (2NF)
Orders PK – order id + item id
Order FKs – customer id, item id
Items PK – item id
Customers PK – customer id

Order id	Order date	Delive			
11024	2024-01-01				
11024	2024-01-01	2024-01-09	Galaxy	1	2201
11083	2025-02-18	2025-02-11	Lenovo	1	3812
11083	2025-02-18	2025-02-11	Boat	1	3812
11083	2025-02-18	2025-02-11	IFB	1	3812
23110	2024-12-23	2024-12-25	Headphone	1	1209

		DOB	Email	address
		1991-08-01	vks@a.com	Bangalore
3812	Jaden	1980-06-05	jdn@b.com	New York
1209	Amit	1994-10-16	amt@c.com	Chennai

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3NF

- Table should be in 1NF and 2NF.
- Non-key attribute should not depend on any other non-key attribute

Order id	Order date	Delivery date	Item Id	Order items	Customer Id
11024	2024-01-01	2024-01-09	100	1	2201
11024	2024-01-01	2024-01-09	200	1	2201
11083	2025-02-18	2025-02-11	300	1	3812
11083	2025-02-18	2025-02-11	400	1	3812
11083	2025-02-18	2025-02-11	500	1	3812
23110	2024-12-23	2024-12-25	600	1	1209

Description	Order price	Item Id
iPhone	999.00	100
Galaxy	900.00	200
Lenovo	250.00	300
Boat	83.00	400
IFB	725.00	500
Headphone	100.00	600

Customer id	name	DOB	Email	address
2201	Vikas	1991-08-01	vks@a.com	Bangalore
3812	Jaden	1980-06-05	jdn@b.com	New York
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3NF

- Table should be in 1NF and 2NF.
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Order id	Order date	Delivery date	Item Id	Order items	Customer Id
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11024	2024-01-01	2024-01-09	200	1	2201
11083	2025-02-18	2025-02-11	300	1	3812
11083	2025-02-18	2025-02-11	400	1	3812
11083	2025-02-18	2025-02-11	500	1	3812
23110	2024-12-23	2024-12-25	600	1	1209

Description	Order price	Item Id
iPhone	999.00	100
Galaxy	900.00	200
Lenovo	250.00	300
Boat	83.00	400
IFB	725.00	500
Headphone	100.00	600

Custom er id	name	DOB	Email	address	Country	State	dob_mm_dd
2201	Vikas	1991-08-01	vks@a.com	Bangalore	India	KA	08-01
3812	Jaden	1980-06-05	jdn@b.com	New York	USA	NY	06-05
1209	Amit	1994-10-16	amt@c.com	Chennai	India	TN	10-16

3NF

Customer id	name	DOB	Email	address	Country	State	dob_mm_dd
2201	Vikas	1991-08-01	vks@a.com	Bangalore	India	KA	08-01
3812	Jaden	1980-06-05	jdn@b.com	New York	USA	NY	06-05
1209	Amit	1994-10-16	amt@c.com	Chennai	India	TN	10-16

Country	State
India	KA
USA	NY
India	TN

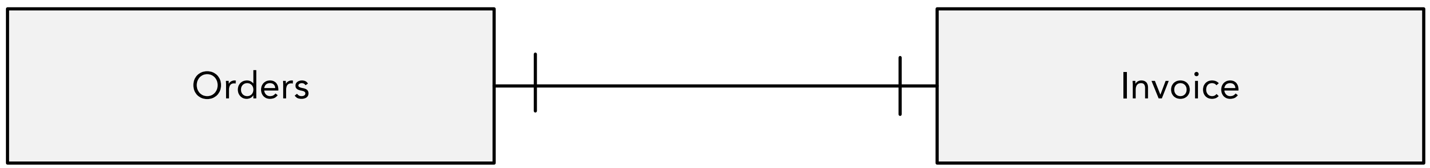
Customer id	name	DOB	Email	address	Country
2201	Vikas	1991-08-01	vks@a.com	Bangalore	India
3812	Jaden	1980-06-05	jdn@b.com	New York	USA
1209	Amit	1994-10-16	amt@c.com	Chennai	India

DOB	dob_mm_dd
1991-08-01	08-01
1980-06-05	06-05
1994-10-16	10-16

Relationships

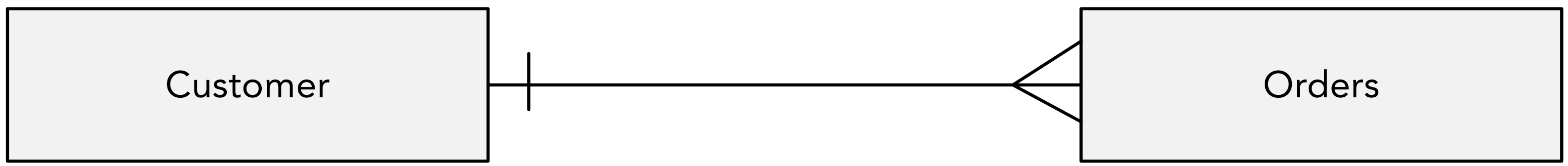
One-to-One (1:1)

- For each record in one entity there will be one corresponding record in the other entity and vice-versa.



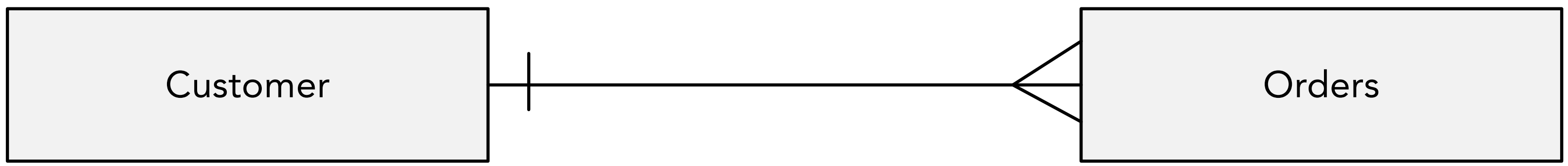
One-to-Many (1:M)

- For each record in one entity there will be many records in the other entity. But not the other way round.



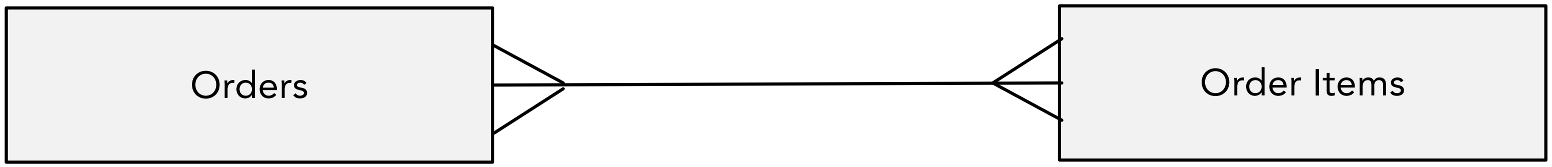
Many-to-One (M:1)

- For multiple records in one entity there will be only one record in the other entity. But not the other way round.



Many-to-Many (M:N)

- For one record in one entity there can be multiple records in the other entity. And vice-versa.



Denormalization

- Opposite of normalization
- Keeping in 1NF mostly.

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11024	2024-01-01	2024-01-09	100	iPhone	999.00	1	2201	Vikas	1991-08-01	vks@a.com	Bangalore
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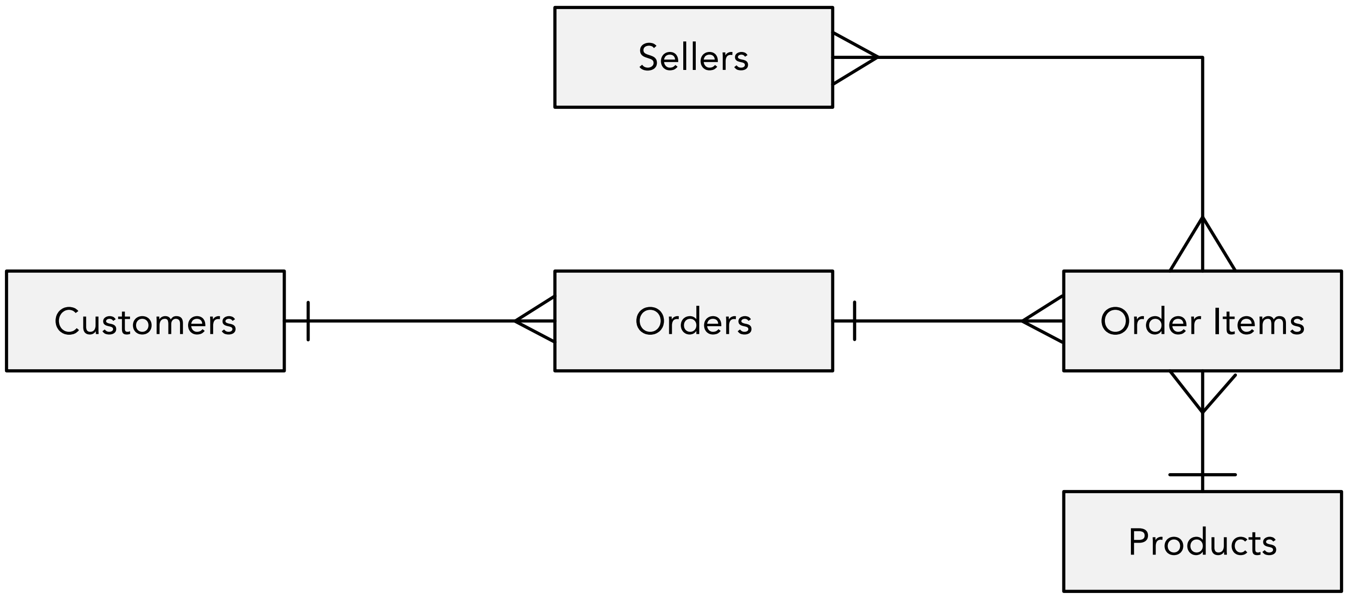
Modelling Techniques

E-R Diagram (OLTP RDBMS)

- Relationship between entities.
- Combination of normalization and (data) relationship.



Dimensional Modelling (OLAP RDBMS)

- Used in Data Wares House (DWH) or Data Lake House.
- Revolves around Facts and Dimensions.
- Facts and Dimensions are also entities.
- Relationships are defined between fact and dimension tables.

Dimensional Modelling

Facts

- Fact table contains numerical performance measurement of the business.
- These are the facts of a business.
- Fact tables should contain numerical data only.
  - Additive – Numerical functions can be applied.
  - Semi-additive – Numerical functions can be applied across some dimensions.
  - Non-additive – Numerical functions cannot be applied at all.
  - Numerical data that are used in calculations.

Order id
11024
11024
11083
23110

Order price	Order items
999.00	1
900.00	1
250.00	1
100.00	1

Dimensions

- Contains description of the business entities.

Order id	Order date	Delivery date	Item id	Description	Order price	Order items	Customer Id	Customer Name	Customer DOB	Customer email	Address
11024	2024-01-01	2024-01-09	100	iPhone	999.00	1	2201	Vikas	1991-08-01	vks@a.com	Bangalore
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Order id	Item Id	Customer Id	Date Key	Order price	Order items
11024	100	2201	1	999.00	1
11024	200	2201	1	900.00	1
11083	300	3812	2	250.00	1
23110	600	1209	3	100.00	1

Item id	Description
100	iPhone
200	Galaxy
300	Lenovo
600	Headphone

Customer Id	Customer Name	Customer DOB	Customer email	Address
2201	Vikas	1991-08-01	vks@a.com	Bangalore
3812	Jaden	1980-06-05	jdn@b.com	New York
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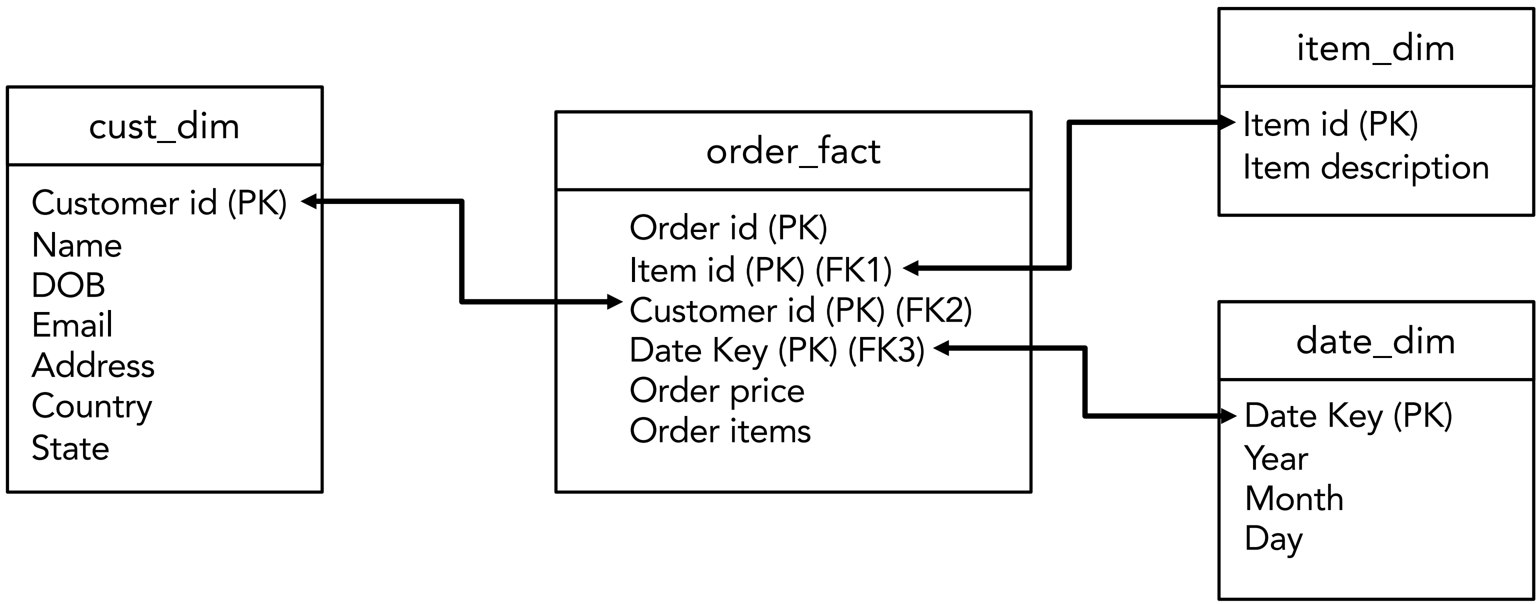
Date Key	Year	Month	day

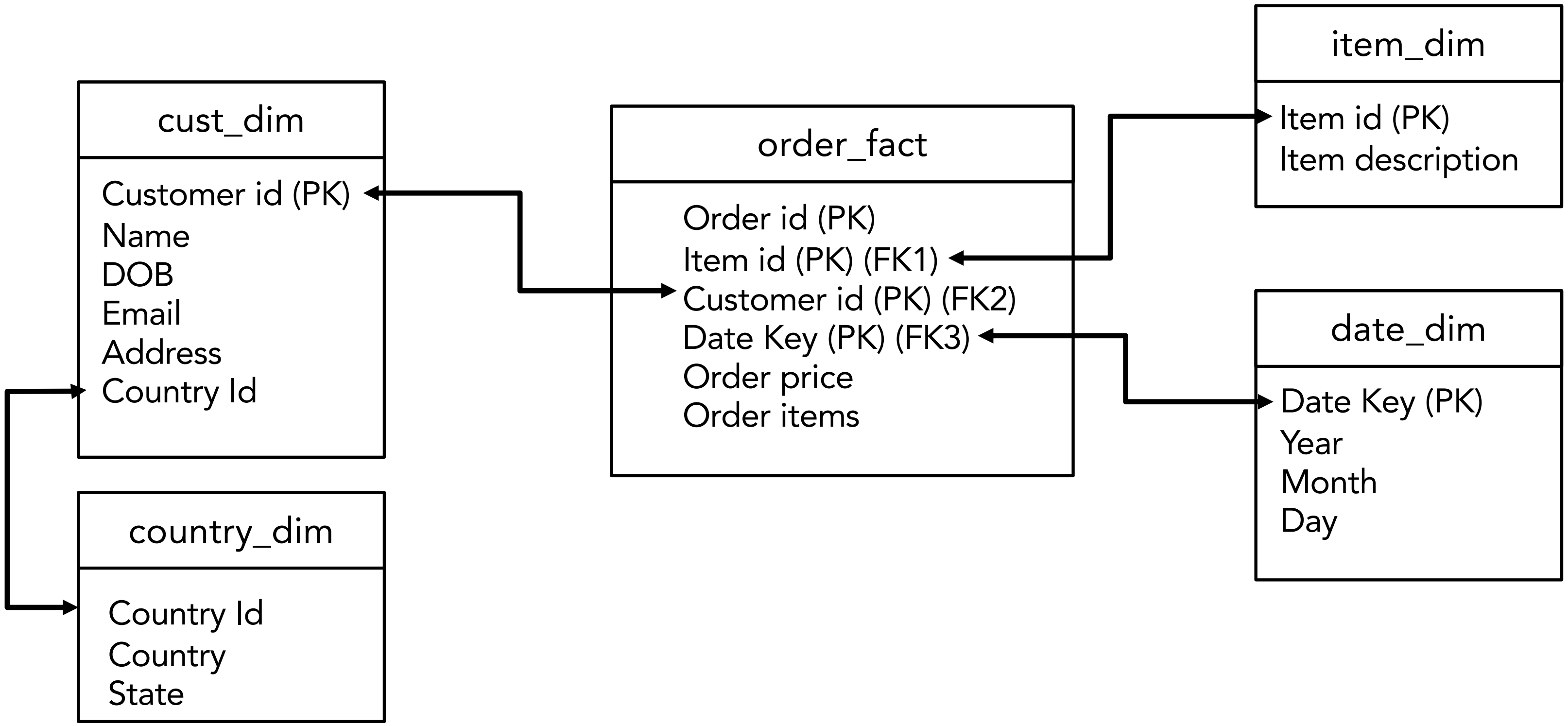
Star Schema

- Fact and dimension tables together.
- Star schema



Snowflake Schema

- Star Schema + 3NF Dimension table



Grains

- Pertain to Fact tables only.
- Level of granularity at which records are stored in Fact tables.
- Combination of columns at which records in a Fact table are unique.
- Level of detail that the data represents in Fact tables.
- Rows that depict the main functionality of business, product, application.
- Primary key of the Fact table.

order_fact
Order id (PK) Item id (PK) (FK1) Customer id (PK) (FK2) Date Key (PK) (FK3) Order price Order items

Grains

- Examples:
  - Retail – One row for each sale.
  - Banking – One row for each transaction.
  - Credit Card – One row for each CC transaction.
  - Insurance – Policy level details.
  - Flight – Departure date and time.
  - School – Admission level.
  - An individual line item on a customer's retail sales ticket as measured by a scanner device
  - An individual transaction against an insurance policy
  - A line item on a bill received from a doctor
  - A boarding pass used by someone on an airplane flight
  - An inventory measurement taken every week for every product in every store.

Hands-on (Identify the Grains)

- A vehicle franchise dealership deals with multiple car brands like GM, Audi, Toyota, Honda, Tata, Mahindra etc. The dealership wants to provide year end offers to increase the sale and clear the inventory. The offers should not impact the profits and targets. Business wants to analyze the data and decide on the type of discounts or deals that can be provided keeping the business profitable. They want to see the trend over the years and months and finalize offers to customers brand-wise.
- An IT company uses an ERP tool to track resource utilization and activity. The tool allows every employee to record the work they do on every opportunity including technology, team, project phase etc. There are various departments in the company. Leadership want to increase the people foreseeing a few big opportunities in the pipeline. The decision to add headcount depends on the month-wise work breakdown across departments over the last 2 years. A DWH system is needed for long term.

Hands-on (Identify the Grains)

- A vehicle franchise dealership deals with multiple car brands like GM, Audi, Toyota, Honda, Tata, Mahindra etc. The dealership wants to provide year end offers to increase the sale and clear the inventory. The offers should not impact the profits and targets. Business wants to analyze the data and decide on the type of discounts or deals that can be provided keeping the business profitable. They want to see the trend over the years and months and finalize offers to customers brand-wise.

Grain - Vehicle sale details to each customer.

- Customer details
- Vehicle details
- Dealership details
- Sale details

Hands-on (Identify the Grains)

- An IT company uses an ERP tool to track resource utilization and activity. The tool allows every employee to record the work they do on every opportunity including technology, team, project phase etc. There are various departments in the company. Leadership want to increase the people foreseeing a few big opportunities in the pipeline. The decision to add headcount depends on the month-wise work breakdown across departments over the last 2 years. A DWH system is needed for long term.

Grain – Record of each employee for each opportunity.

- Employee details
- Opportunity details
- Project details
- Department details
- Record of employees working on opportunities.

Date Dimension

- Contains all possible details on date. Built in advance for 10 yrs +/- 5 yrs.

- Date key
- Date
- Day of Week
- Day of Month
- Day of Year
- Day of fiscal month
- Day of fiscal year
- Day no. in epoch
- Week no. in epoch
- Month no. in epoch
- Last day in week
- Last day in month
- Week ending date
- YYYY-MM
- Year quarter
- Fiscal week etc.

Dimensional Modelling Steps

- Understand the business process.
- Define Grain.
- Define dimension tables.
- Define fact tables.

Example - Retail Store

- Customers
- Orders
- Order Items
- Products
- Sellers

ER Diagram

- Customers
- Orders
- Order Items
- Products
- Sellers

Customers

Column name	Data type
customer_id	bigint
customer_name	string
company	string
shipping_address	string
zipcode	bigint
city	string
state	string
country	string
dob	string
email	string
gender	string
nationality	string
phone	string
username	string
credit_card_no	string
credit_card_type	string
paypal_account	string

Orders

Column name	Data type
order_id	bigint
customer_id	bigint
seller_id	bigint
order_date	string
payment_method	string

Order Items

Column name	Data type
order_id	bigint
product_id	bigint

Sellers

Column name	Data type
seller_id	bigint
seller_address	string
seller_state	string
seller_zipcode	bigint
seller_country	string
seller_rating	bigint
seller_active_date	string
seller_business_type	string
seller_company_name	string

Products

Column name	Data type
product_id	bigint
product_name	string
price	double
discount	double

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DWH Requirement

- Inventory management – Based on the products sold across region (zip code, city, state) inventory should be planned for each seller.
- Customer offer – Based on the most used payment method (credit card, debit card, visa etc.), EMI options would be given along with cashback offer.
- Seller offer – Based on no. of products sold and rating sellers will be given discount on raw products (b2b).
- Trend of selling – Find out the trend of product sales over the months in a year. Will be used in manufacture plants for planning.

Define Grains 1

- Inventory management – Based on the products sold across region (zip code, city, state) inventory should be planned for each seller.

Column name	Data type
customer_id	bigint
customer_name	string
company	string
shipping_address	string
zipcode	bigint
city	string
state	string
country	string
dob	string
email	string
gender	string
nationality	string
phone	string
username	string
credit_card_no	string
credit_card_type	string
paypal_account	string

Column name	Data type
order_id	bigint
customer_id	bigint
seller_id	bigint
order_date	string
payment_method	string

Column name	Data type
order_id	bigint
product_id	bigint

Column name	Data type
product_id	bigint
product_name	string
price	double
discount	double

Column name	Data type
seller_id	bigint
seller_address	string
seller_state	string
seller_zipcode	bigint
seller_country	string
seller_rating	bigint
seller_active_date	string
seller_business_type	string
seller_company_name	string

Grain – Products / Order Items

Define Grains 2

- Inventory management – Based on the products sold across region (zip code, city, state) inventory should be planned for each seller.

Column name	Data type
customer_id	bigint
customer_name	string
company	string
shipping_address	string
zipcode	bigint
city	string
state	string
country	string
dob	string
email	string
gender	string
nationality	string
phone	string
username	string
credit_card_no	string
credit_card_type	string
paypal_account	string

Column name	Data type
order_id	bigint
customer_id	bigint
seller_id	bigint
order_date	string
payment_method	string

Column name	Data type
order_id	bigint
product_id	bigint

Column name	Data type
product_id	bigint
product_name	string
price	double
discount	double

Column name	Data type
seller_id	bigint
seller_address	string
seller_state	string
seller_zipcode	bigint
seller_country	string
seller_rating	bigint
seller_active_date	string
seller_business_type	string
seller_company_name	string

Grain – Products / Order Items

Define Dimension Tables

- Date
- Customer
- Seller
- Product
- Payment

date_dim

date_key (PK)
date
day
month
year
dow
doy
dom

product_dim

product_id (PK)
product_name

customer_dim

customer_id (PK)	bigint
customer_name	string
company	string
shipping_address	string
zipcode	bigint
city	string
state	string
country	string
dob	string
email	string
gender	string
nationality	string
phone	string
username	string
credit_card_no	string
credit_card_type	string
paypal_account	string

seller_dim

seller_id (PK)	bigint
seller_address	string
seller_state	string
seller_zipcode	bigint
seller_country	string
seller_rating	bigint
seller_active_date	string
seller_business_type	string
seller_company_name	string

payment_dim

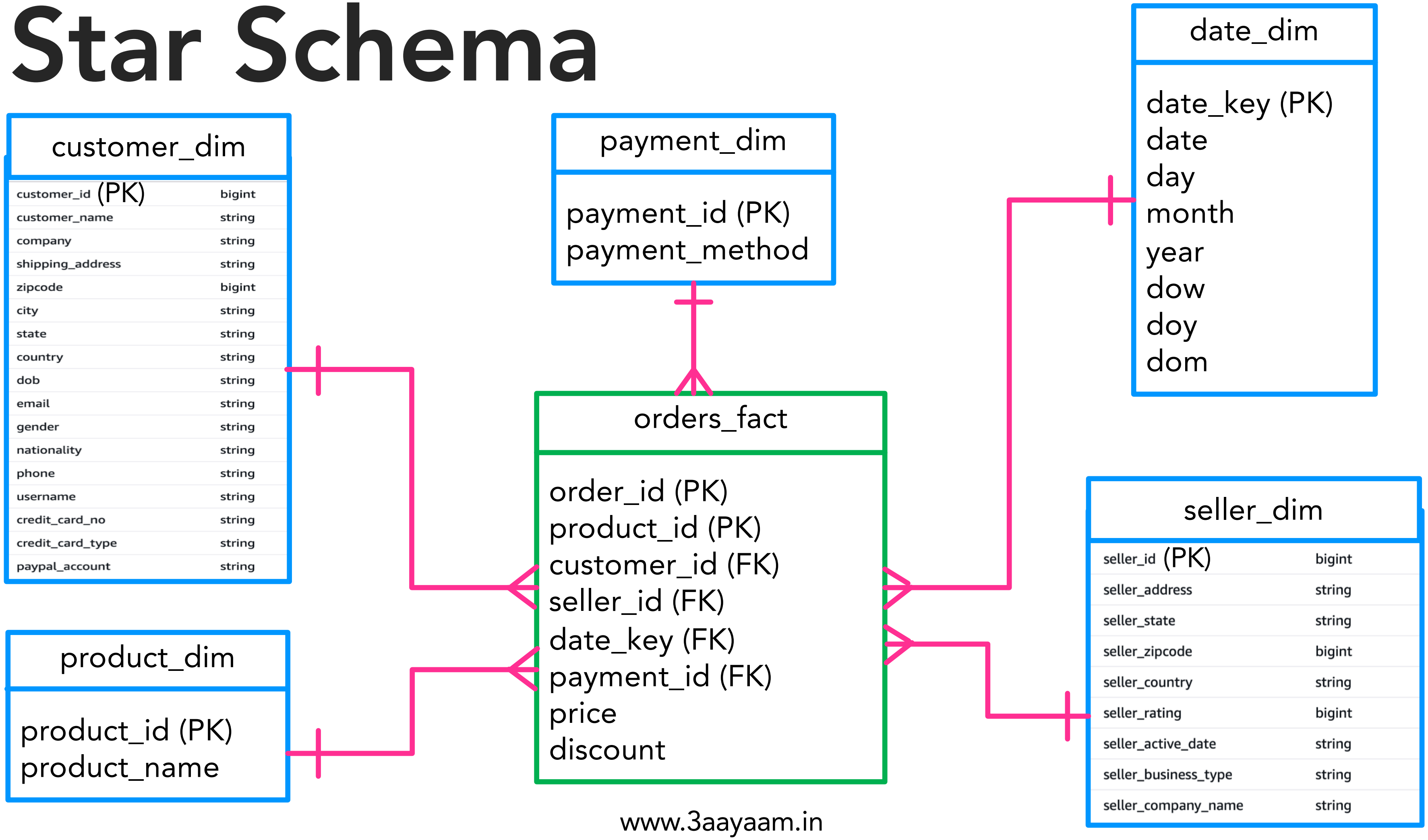
payment_id (PK)
payment_method

Define Fact Tables

- Orders (with individual products as line items)
  - Order id (PK)
  - Product id (PK)
  - Price
  - Discount

orders_fact
order_id (PK) product_id (PK) customer_id (FK) seller_id (FK) date_key (FK) payment_id (FK) price discount

Star Schema



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Types of Fact Tables

- Transactional fact tables. Grain is at individual transaction level.
- Periodic snapshot facts.
  - Extended version of Transactional Fact tables that captures summary over a period.
  - Daily, weekly, monthly intervals.
  - Grain is at the period level.
- Accumulating snapshot facts.
  - Contains accumulated data over a period of time.
  - Aggregated data from beginning till the date when snapshot is stored.
- Conformed dimensions – Dimension tables that are used across multiple star schemas.
- Conformed facts – Fact tables that use conformed dimension tables. Multiple fact tables in a single star schema.

Types of Fact Tables

- Additive facts – Can be summed across all dimensions.
  - Price, quantity.
- Semi-additive facts – Can be summed across only a few dimensions.
  - Orders per day for each item.
- Non-additive facts – Cannot be summed at all.
  - Discount.

Slowly Changing Dimensions (SCD)

- SCD1
  - Overwrite the old attribute value with current value.
  - The attribute always contains the recent value.
- SCD2
  - A new row is added with the changed value.
  - Dimension table contains historical data of all changes.
- SCD3
  - A new column is added with the changed value.

Cust id	Name	Address	Country	Eff Date
1100	Vikas	BLR	India	2010-01-01
9828	John	NY	USA	2004-01-01

SCD 1

Cust id	Name	Address	Country	Eff Date
1100	Vikas	MUM	India	2025-01-01
9828	John	NY	USA	2004-01-01

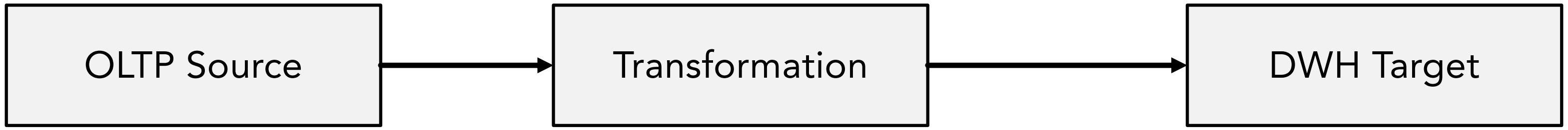
SCD 2

Cust id	Name	Address	Country	Eff Date
1100	Vikas	BLR	India	2010-01-01
1100	Vikas	MUM	India	2025-01-01
9828	John	NY	USA	2004-01-01

SCD 3

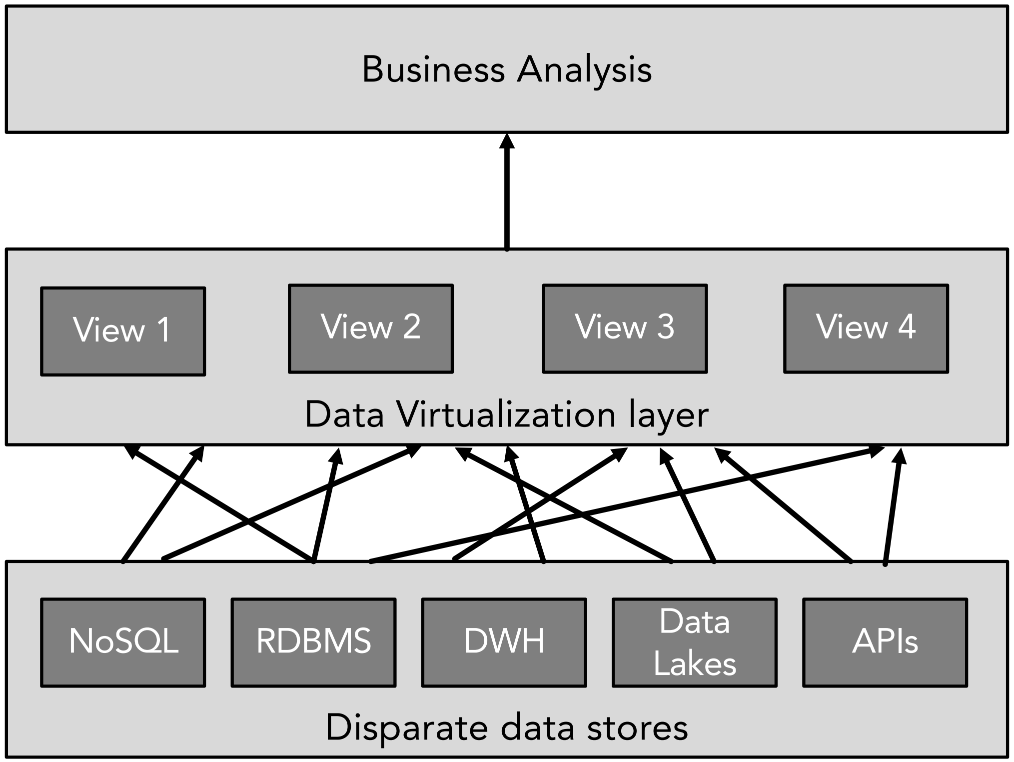
Cust id	Name	Prior 1 addr	Curr addr	Country	Prior 1 Eff Date	Curr Eff Date
1100	Vikas	BLR	MUM	India	2010-01-01	2025-01-01
9828	John	NY	NA	USA	2004-01-01	NA

Transformation (OLTP to DWH)



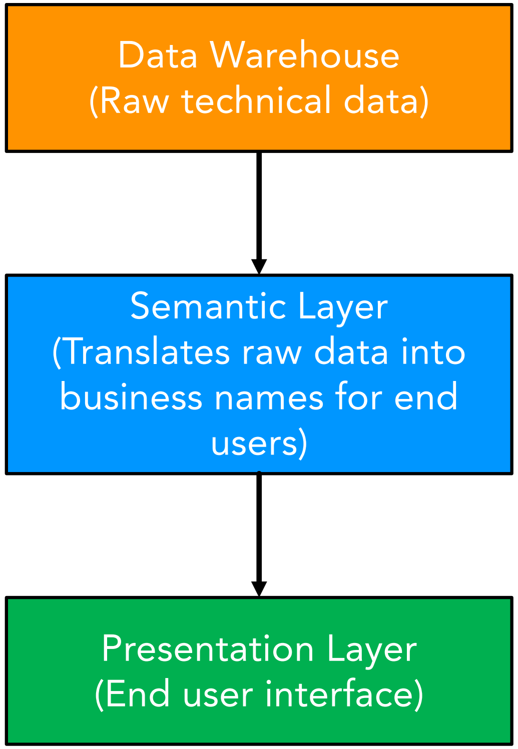
- Data from sources are moved to target after transformation.
- ETL or ELT process.
  - Spark, AWS Glue, AWS EMR, GCP Dataproc, Redshift (ELT), Showflake (ELT), BigQuery (ELT).
  - Spark streaming, Flink, Redshift streaming ingestion, BigQuery, Snowflake Snowpipe etc.

Data Virtualization



- Disparate data sources – SQL, NoSQL, DWH, Data APIs, Data Lakes, Lakehouse etc.
- How to use these data together for business analysis?
- Data Virtualization technique.
  - Denodo, Athena, Presto, Trino.
- Business analysis.
  - PowerBI. Tableau, QuickSight etc.

Semantic & Presentation Layers



- Data in warehouse is considered to be raw (from the perspective of end users – CxOs, BDs, Bas etc.)
- Semantic layer is used to translate the raw data into business friendly names and data (joining, renaming) so that end users can understand them.
  - PowerBI, Tableau, Dremio, Qlik etc.
- Presentation layer is the interface that is used by end users.
  - PowerBI, Tableau, Dremio, Qlik etc.

Summary

- Data modelling intro & terminologies (entity, attribute, relation, PK, FK)
- Modelling Concepts (Normalization, 1NF, 2NF, 3NF).
- Relationships (based on data – 1:1, 1:m, m:1, m:n). Denormalization.
- ER diagram, Dimensional modelling.
- Facts, Dimensions, Star & Snowflake schema.
- Grains.
- Types of facts and dimension tables.
- Data virtualization
- Semantic & Presentation layers.

Thank You!

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